Genes simulations – y10 **Name** _____

Activity 1: Intro to Genes. Go to http://www.scootle.edu.au/ec/viewing/L5918/index.html and click 'student login' (top left). Enter the code **SSQTTB.**

As you go through, fill in the blanks and answer these questions:

1. All living things are _____
2. Some organisms only have _____ but most have _____
3. The nucleus is _____
4. Chromosomes carry _____ information.
5. Are the number of chromosomes the same for each species? yes / no
6. How many chromosomes do humans have? _____
7. DNA stands for _____
8. DNA is a _____ made up of building blocks called _____. DNA is in the shape of a _____. The sequence of nucleotides determines the _____ of an organism.
9. Genes carry the instructions for _____ of an organism.
10. DNA _____ are composed of _____ of _____.

Activity 2: What is DNA. Go to https://www.scootle.edu.au/ec/viewing/L5919/index.html and click 'student login' (top left). Enter the code **SSQTTB.**

As you go through, fill in the blanks and answer these questions:

1. DNA is the _____ material of all living things.
2. Can you see DNA with a normal classroom microscope? yes / no
3. In the 1800s scientists discovered _____. They looked like _____ in the _____ of each cell.
4. Later, scientists discovered _____ _____ which can exist in the form of chromosomes. This is what we now call _____.
5. What 5 elements are in DNA?

   _____

   _____
6. What 6 components (made of those elements) are in DNA? (press 'play')

   _____

   _____
7. In the 1950s, scientists used the process of _____ to find that DNA had the shape of a _____ _____ - a sort of _____ _____.
8. Which two components are in the backbone?

   _____
9. Which bases pair up with which?

   _____
10. Bases are held together with

    _____

11. A unit formed by a _____, a _____ and it's _____ is called a _____.

12. When you form base pairs, the strand that you are building is called the _____.

13. Human chromosomes can have from _____ genes up to _____.

When you have finished, go to https://educationalgames.nobelprize.org/educational/medicine/dna_double_helix/dnahelix.html and play the DNA game.